LILONGWE UNIVERSITY
OF AGRICULTURE & NATURAL
RESOURCES (LUANAR)

Knowlege, Innovation, Excellence

BIOLOGY 2

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BIOLOGY 2

- Course Code; NBIO21202

Aim

- To develop an understanding and skills in life sciences as applied to agriculture, environment and natural resource management

Learning outcomes

a. describe Mendelian genetics

b. describe the classification system of plants into the different taxa.

c. use the classification system of plants to describe the different taxa of plants

Topics

- **Topic One: Genetics and evolution**
  - Mendelian genetics
  - Population genetics
  - Molecular genetics/neo-mendelian genetics
  - Concept of evolution
  - Evidence and mechanisms of evolution
  - Theories of the origin of life

Topic Two: Botany

- Plant organs: leaves, stems, roots
- Plant tissues: epidermal tissue, ground tissue, vascular tissue
- Classification of Plants:
- Tropism and photoperiodism
- Reproduction in plants
- Phytochromes and plant hormones
- Water and Nutrient uptake;

Topic Three: Ecology of Plants and Animals

- The physical and biotic environment of organisms
- The ecological niches, habitats, communities and ecosystems
- The food chains and Energy Pyramids
- The growth and regulation of populations
- Sampling methods in ecology
- Agricultural and non-agricultural ecosystems

Practicals

a. Genetics

- Study of models on DNA and RNA structures
- Demonstrations on monohybrid, dihybrid, trihybrid, test crossing and back crossing

Practical cont..

b. Botany

- Demonstration of plant taxa and use of keys for plant identification
- Observing the morphology of leaves, stems, roots, and flowers
- Seed germination
- Test for starch,
- Demonstration of osmosis, osmotic pressure, water uptake and transpiration in plants

Practical cont..

a. Ecology of animals and plants

- Sampling methods in ecology
- Use of quadrats and transects
- Finding the probability and Chi-square test Observing
- comparing the agricultural and non-agricultural ecosystems around NRC

Prescribed Texts

Elrod, S.L. & Stansfield, W.D. (2010), Theory and problems of genetics, 5th Edition, Schaum's Outline of Genetics. McGraw Hill Education

Mauseth, J.D. (2014), Botany, Burlington, Massachusetts, Jones and Bartlett Publishers

Solomon, E. P., Berg, L. R., & Martin, D. W. (2011). Biology 9th Edition. Belmont, CA: Brooks/Cole

Recommended Texts

Alberts B., Bray D., Hopkin K., Johnson A., Lewis J., Raff M., Roberts K., Waiter P. (2014). Essential Cell Biology, 4th Edition, Garland Science, Taylor & Francis Group, LLC.

Hartl, D.L., and Jones, E.W., (2006). Essential Genetics- A Genomic Perspective. 4th ed. Sudbury MA, Jones and Bartlett Publishers.

Klug, W.S., and Cummings, M.R., Spencer, C.A., & Palladino, M.A., (2014). Essentials of Genetics. 8th Edition. New Jersey: Pearson

Raven P. et al., (2020). Biology, 12th Edition, McGraw-Hill

Topic one: Genetics and Evolution.

- **a. Genetics Terms and definitions**
- **1. What Is gene?**

  1. **Classical definition**: A gene is a discrete unit of heredity that is passed down from parents to offspring and determines a specific trait or characteristic.

  2. **Genetic definition**: A gene is a unit of inheritance that occupies a specific location on a chromosome and determines a particular characteristic or trait.

Gene meaning

- 3. **Molecular definition**: A gene is a segment of DNA (deoxyribonucleic acid) that codes for a specific protein or functional product.

- 4. **Functional definition**: A gene is a sequence of DNA that is transcribed into RNA (ribonucleic acid) and translated into a protein, which performs a specific function in the organism.

- 5. Genomic definition: A gene is a region of DNA that is **encoded** in the genome of an organism and provides the instructions for the development, growth, and function of that organism.

Key aspects of genes

- Gene Expression: The process by which a gene's information is converted into a functional product.-
- Regulation: The control of gene expression in response to environmental or cellular cues.-
- Gene Mutation: Changes in the DNA sequence of a gene that can affect its function or expression.-
- Inheritance: The passing down of genes from parents to offspring.

Where are genes stored in a cell?

- In eukaryotes (such as animals, plants, and fungi), genes are contained chromosomes located within the cell nucleus.
- The **mitochondria** (in animals) and the **chloroplasts** (in plants) and **ribosomes** also contain small subsets of genes distinct from the genes found in the nucleus.
- In prokaryotes (organisms lacking a distinct nucleus, such as bacteria), genes are contained in a single chromosome that is free-floating in the cell cytoplasm.
- Many bacteria also contain **plasmids**—extrachromosomal genetic elements with a small number of genes.

- **What is a locus?**
- a locus (**plural: loci)** refers to a specific location on a chromosome where a particular gene is found.
- It's essentially a precise address on the chromosome that identifies the position of a gene or a DNA sequence.

- Each locus can have different variants or **alleles**, which are alternative forms of the gene or DNA sequence at that specific location.
- These alleles can influence various **traits or characteristics**, such as eye color, height, or even susceptibility to certain diseases.

What is Genotype?

- a genotype refers to the complete set of genes that an individual organism inherits from its parents.
- It's the genetic makeup of an individual, written in the DNA sequence of their chromosomes.
- An individual's genotype is made up of two copies of each gene, one inherited from each parent

genotype

- The genotype determines various aspects of an individual's traits and characteristics, such as:-
- Physical characteristics (eye color, hair color, height, etc.)-
- Behavioral characteristics
- Susceptibility or resistance to certain diseases-
- Response to environmental factors

What is a phenotype?

- **Also called character or trait**
- a phenotype refers to the physical and behavioral characteristics of an individual that result from the interaction of their **genotype** and the **environment**.
- In other words, phenotype is the outward expression of genes, influenced by environmental factors.

Phenotype is a result of complex interactions

- 1. Genotype (the genetic information)
- 2. Environmental factors (diet, temperature, exposure to toxins, etc.)
- 3. Epigenetic factors (gene expression regulation)

Epigenetic factors?

- Epigenetic factors are elements that influence gene expression without altering the DNA sequence itself.
- They play a crucial role in shaping an individual's phenotype, development, and adaptation to the environment.

Some Examples of epigenetic factors

- 1. **DNA methylation**: chemical modifications to DNA that silence gene expression
- 2. **Histone modification**: changes to histone proteins that DNA wraps around, affecting gene accessibility
- 3. **Chromatin remodeling**: reorganization of chromatin structure to allow or block gene expression

SOME EXAMPLES OF EPIGENETIC FACTORS

- 4. **Environmental factors**:
- 5. Exposure to **toxins** and **pollutants** -
- 6. **Stress and psychological factors** - Social and cultural influences
- 7. **Hormonal** and **metabolic** factors: - Hormone levels and balance

What is genetics?

- Genetics is the scientific study of **heredity, variation**, and the **transmission** of traits from parents to offspring.
- It's a branch of biology that explores how genes, genetic information, and environmental factors interact to shape the characteristics and behaviors of living organisms.

Branches of Genetics

- 1. **Mendelian genetics**: studies the inheritance patterns of traits and genes
- 2. **Molecular genetics**: examines the structure, function, and regulation of genes at the molecular level
- 3**. Population genetics**: explores the distribution and frequency of genetic variations within populations

Branches

- 4. **Quantitative genetics**: analyzes the contribution of multiple genes to complex traits
- 5. **Epigenetics**: investigates gene expression and regulation beyond the DNA sequence
- 6. **Genomics:** focuses on the study of entire genomes, including their structure, function, and evolution

Application of genetics

- 1. Medicine (genetic counseling, personalized medicine)
- 2. Agriculture (crop improvement, animal breeding)
- 3. Biotechnology (genetic engineering, gene editing)
- 4. Forensic science (DNA fingerprinting, identification)
- 5. Anthropology (human evolution, population studies)

Mendelian genetics

- **Historical puzzle of inheritance**
- **Artificial selection has been important since recorded history.**
  - **selection of animals for domestication and**
  - **selection of plants for cultivation.**

Breeders' Puzzle…..

19th century – Breeders desire precise techniques for controlled matings in plants and animals in order to produce desired traits in many of offspring

BREEDERS' PUZZLE…..

- Breeders could not explain why traits would sometimes disappear and then reappear in subsequent generations.

Animal Breeders mystery

- Wanted this.

Animal breeders

- Got this.

Gregor Mendel, the father of modern genetics

- Mendelian genetics, named after Gregor Johann Mendel.
- Mendel, an Austrian monk and scientist, conducted experiments on pea plants in the 1860s, uncovering the fundamental principles of inheritance.

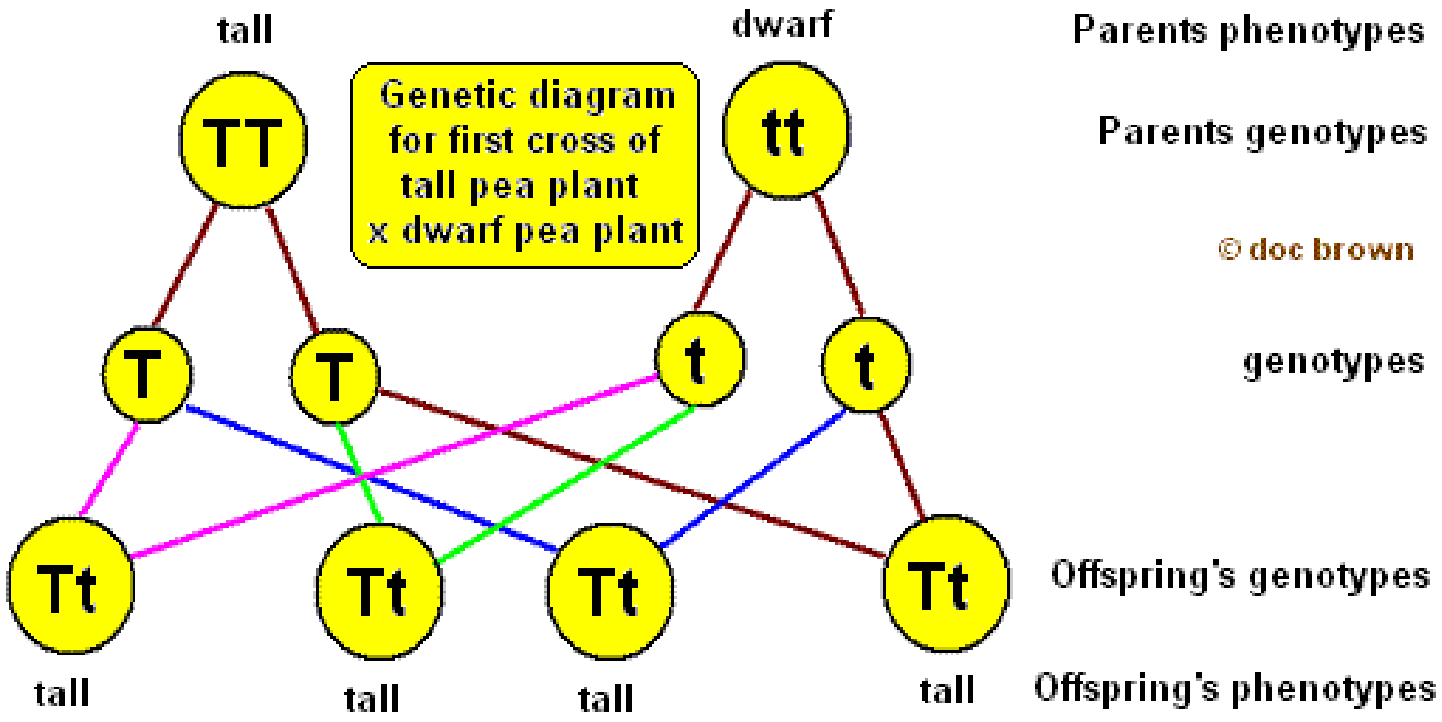
Mendel's Work (1860s):

- - Conducted experiments on pea plants (Pisum sativum)
- - Studied traits like height, flower color, and seed shape
- - Mendel studied peas with different phenotypes using genetic crosses.

What is a genetic cross?

- **A genetic cross** is the purposeful mating of two individuals resulting in the combination of genetic material in the offspring.
- **Cross** is a flow diagram which shows how alleles of the same gene are passed from parents to offsprings.
- Example a genetic cross between two pea plants one tall and the other dwarf.
- Allele for tall dominant to allele for dwarf.

Example of genetic cross



INFORMATION OUTLINED IN THE GENETIC CROSS

- **1. Parental generation:** A pair of parents used in breeding.
- 2. **Purebred/Homozygous:** A condition whereby two similar alleles are present at a locus in diploid cells**.**
- **3. Homozygous dominant or homozygous recessive ???**
- **4. True breeding:** organisms that are homozygous for a particular gene

- **5: Heterozygous:** a state in which two contrasting alleles of the same gene are present in diploid state.
- **Heterozygous can be inherited in three ways:**
  - **1. Complete dominance:** One allele masks the expression of the other completely.
  - **2. Codominance:** The expression of both alleles in heterozygous. Example the AB blood group.

- **3. Incomplete dominance:** An allele is not dominant completely on the other as a result, an intermediate phenotype can be seen in the individual.

- For example, Pink colour flowers are seen in Snapdragon, one allele for being red and the other for white.

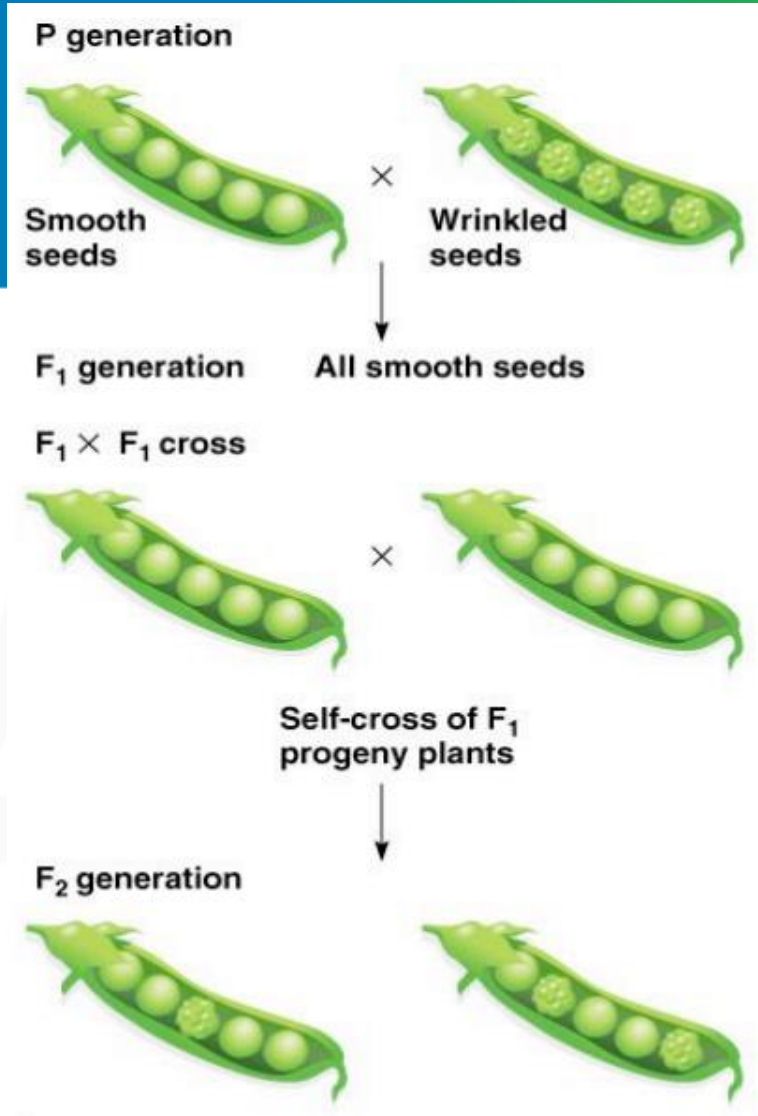
- **6. F1 Generation:** Offsprings born from P generation
- **7. F2 Generation:** offsprings born from inbreeding F1 generation

Monohybrid crosses and Mendel's Principle of segregation

- A cross used to study a singe trait/phenotype with two variations.
- Example flower colour is a phenotype and purple and white flowers are two variations.

Example:

- Mendel crossed two true-breeding parents one with smooth seed and the other with wrinkled seed.
- The allele for smooth seeds is dominant to allele for wrinkled seed.
- Analyze the test cross up to F2 Generation


P generation
Smooth seeds
×
Wrinkled seeds
F1 generation
All smooth seeds
F1 × F1 cross
×
Self-cross of F1 progeny plants
F2 generation

From this cross, Mendel proposed the law of segregation:

- Which states that:

Mendel's first law states that each phenotype is determined by a pair of factors which we now call genes. These pair of factors segregate from each other while forming gametes. At the time of fertilization, the segregated gametes pair again reunite to form new combination.

Punnett square

- A Punnett square is a grid that enables one to predict the outcome of a simple genetic crosses

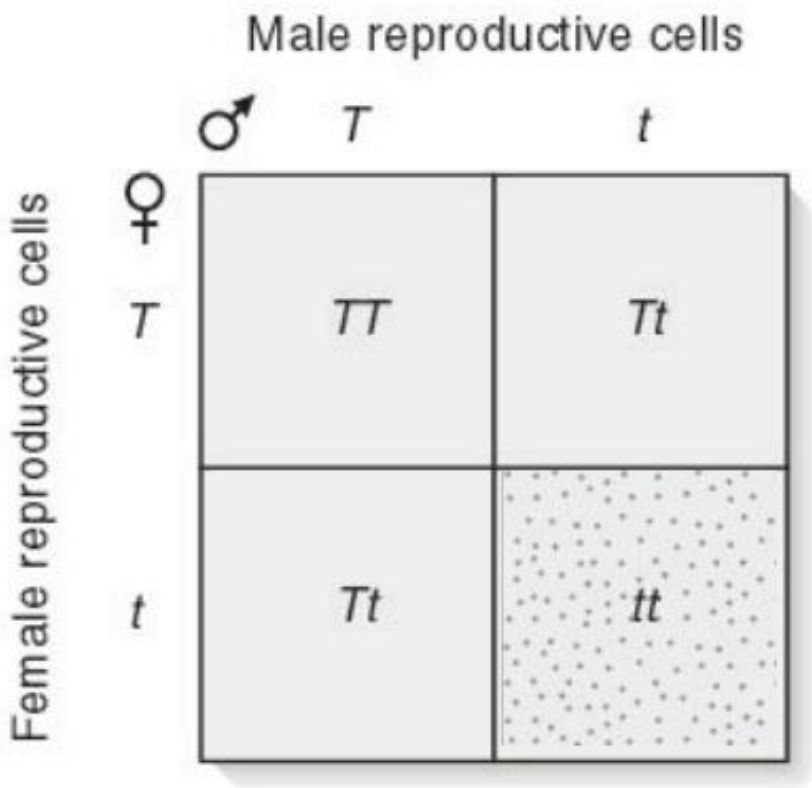
- Proposed by English geneticist Reginald Punnett

How to use Punnett square

- 1. Write down the genotypes of both parents
  - Male parent = *Tt*
  - Female parent = *Tt*

- 2. Write down the possible gametes each parent can make.
  - Male gametes: *T* or *t*
  - Female gametes: *T* or *t*

- 3. Create an empty Punnett square



- 4. Fill in the Punnett square with the possible genotypes of the offspring

- 5. Determine the relative proportions of genotypes and phenotypes of the offspring

  - Genotypic ratio
    - TT : Tt : tt
    - 1 : 2 : 1

  - Phenotypic ratio
    - Tall : dwarf
    - 3 : 1

Calculations using genotypic and phenotypic ratios

- 1. In a cross between two plants with the genotype “Tt", what is the expected genotypic and phenotypic ratios of the offspring, if T codes for Tall plant and t for dwarf plant and allele for tall dominant to dwarf.
- 2. In a study on pea seed shape, the F2 generation produced had 219 plants with smooth seed and 73 plants with wrinkled seed. What is the genotypes of F2 and P generations.

Test cross:

- A test cross is a breeding experiment used to determine the genotype of an individual with a dominant trait.
- It involves crossing the individual with a recessive homozygote.

Example

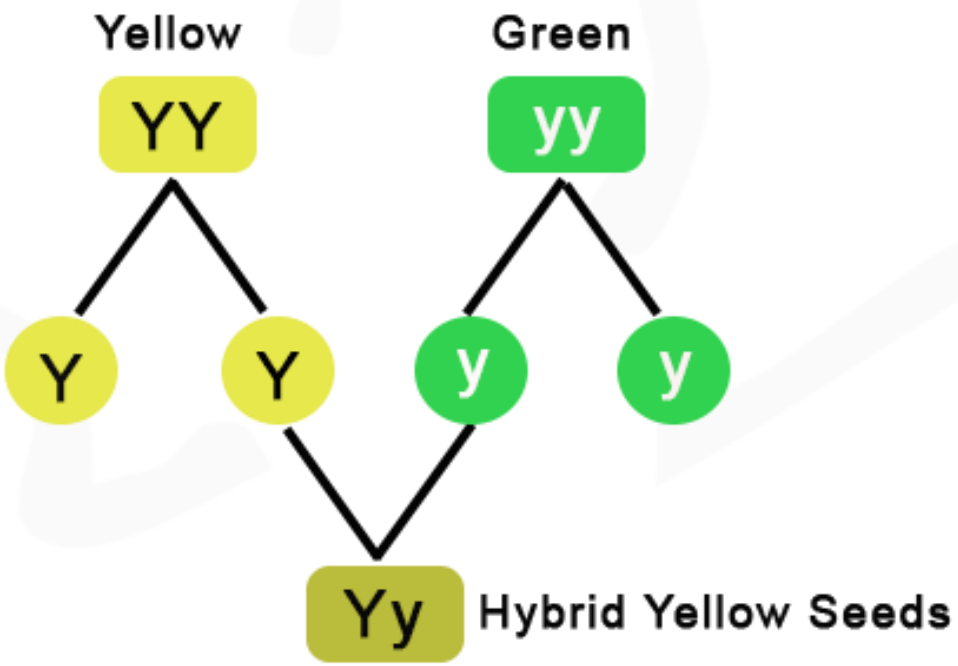
- In pea plants the gene for seed colour has two alleles. One for yellow colour and one for green colour.
- Yellow is dominant to green.
- Let ‘Y’ to denote allele for yellow seed and ‘y’ to denote allele for green.

- The question now is: is the genotype of a yellow seed ‘YY’ (homozygous) or ‘Yy’ Heterozygous?.

- Perform a test cross with a known homozygous recessive (yy).
- This must be done in the field by growing the plants.

Expected results

- A. If the yellow seed was homozygous



b. If the yellow seed was heterozygous

	y	y
Y	Yy	Yy
y	yy	yy

Dihybrid cross and Mendel's law of independent assortment

- What happens when more than one pair of characters is simultaneously involved in the crosses?

- In one dihybrid cross experiment, Mendel studied the inheritance of seed color and seed shape.

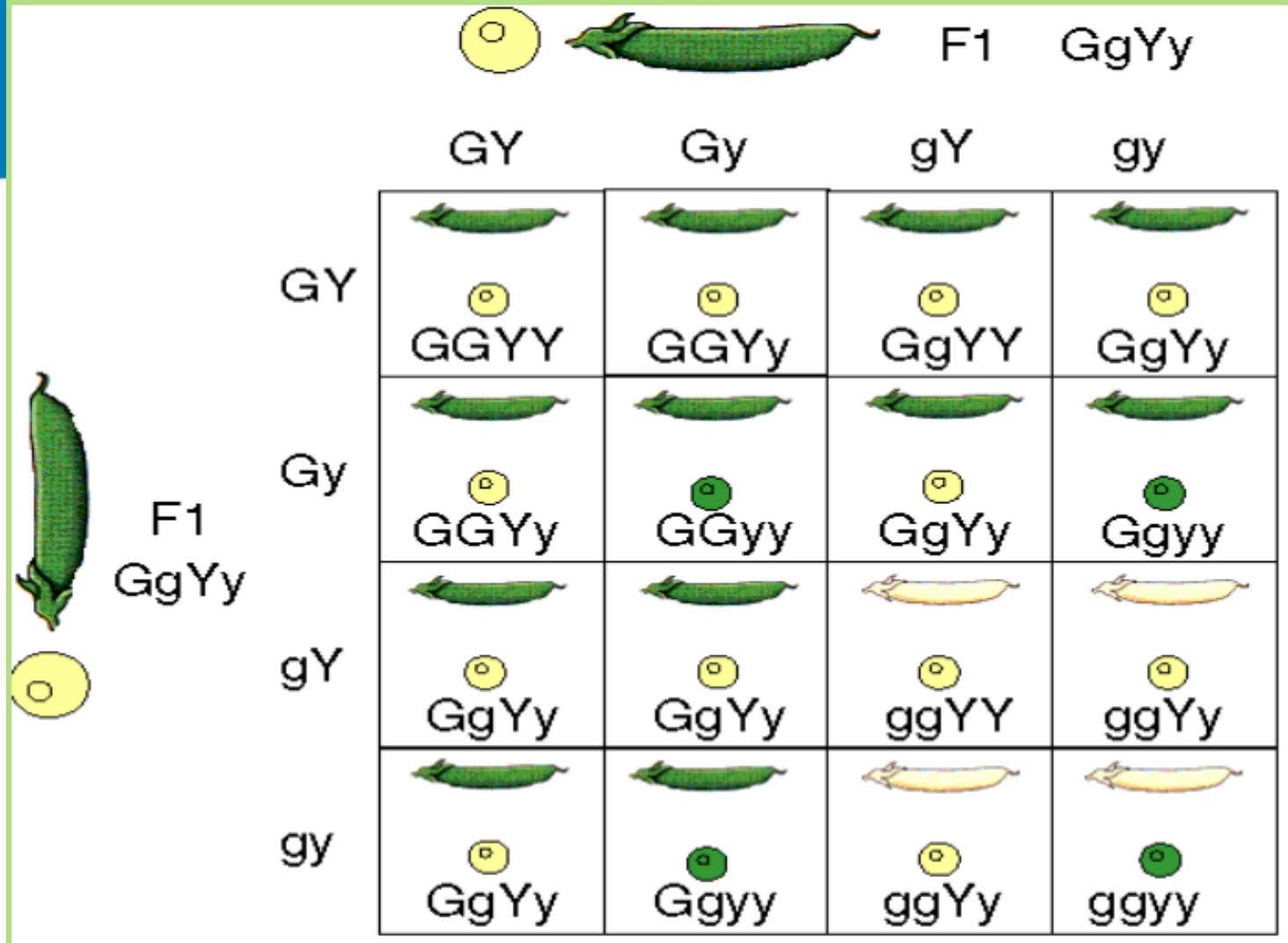
- The allele for yellow seeds (Y) is dominant to the allele for green seeds (y).
- The allele for round seeds (R) is dominant to the allele for wrinkled seeds (r).

- Mendel crossed true-breeding plants that had yellow, round seeds (YYRR) with true-breeding plants that has green, wrinkled seeds (yyrr).

- The resulting F1 dihybrid progeny were heterozygous for both traits (RrYy) and had round yellow seeds, the dominant phenotypes.

- From the F1 generation, Mendel could not tell if the two characters were inherited independently or not, so he allowed the F1 progeny to self-pollinate.
- That is **RrYy X RrYy.**
- **Let's use a Punnett square with 16 boxes to solve this.**

Dihybrid cross



Mendel's observations and conclusion.

- Alleles of different genes were inherited independently

- He discovered that the combinations of traits in the offspring of his crosses did not always match the combinations of traits in the parental organisms

Mendel's law of independent assortment states

- **The alleles of two (or more) different genes get sorted into gametes independently of one another.**

- **In other words, the allele a gamete receives for one gene does not influence the allele received for another gene.**

Difference between law of segregation and independent assortment

- The law of segregation states that every individual possesses two alleles and only one allele is passed on to the offspring.
- The law of independent assortment states that the inheritance of one pair of genes is independent of inheritance of another pair.

Mendel's law of dominance

- One allele can be dominant over another allele, resulting in the dominant allele being expressed in the phenotype.
- The recessive allele will only be expressed if an individual is homozygous recessive (has two copies of the recessive allele).

Exceptions to Mendel's laws

- Through the course of time, geneticists have discovered some observations that could not be explained by Mendel's principles.
- They are referred to as **deviations/exceptions** to Mendel's work.

Reasons for deviations

- Mendel observed complete dominant genes where one allele masks the expression of the other.
- Mendel did experiments on genes which are on different chromosomes and can segregate easily.

Exceptions

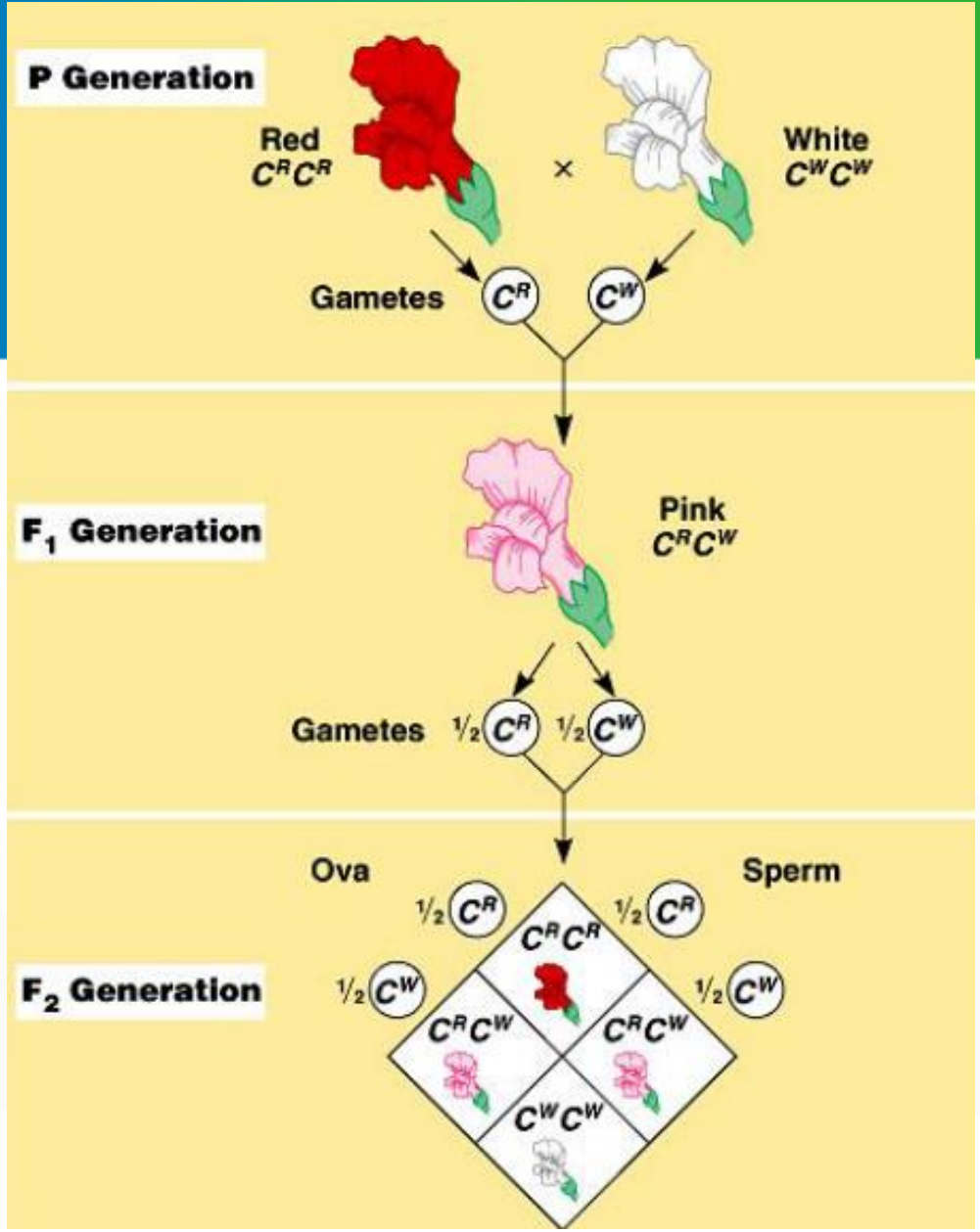
- 1. Incomplete dominance
- 2. Codominance:
- 3. Pleiotropy:

- 4. Polygenic inheritance:
- 5. Gene-environment interaction:
- 6. Linkage:
- 8. Lethal genes

1. Incomplete dominance

- When two different alleles of the same gene produce a blended phenotype
- Eg flower colour in a plant called *Mirabilis jalapa*
- Alleles produce red and white flowers

- A cross between Red and white and pink F1 generation


P Generation
Red
C^RC^R
×
White
C^WC^W
Gametes
C^R
C^W
F_1 Generation
Pink
C^RC^W
Gametes
½ C^R
½ C^W
F_2 Generation
Ova
Sperm
½ C^R
½ C^R
½ C^W
½ C^W
C^RC^R
C^RC^W
C^RC^W
C^WC^W

results

- Phenotypic ratios of F2 generation
- Red : Pink : White are 1:2:1 not 3:1 ration as in normal monohybrid cross of heterozygous individuals.

- Note, C stands for Colour

2. Codominance

- *When both alleles have an equal effect on the phenotype, resulting in a combination of both traits*
- The example is AB blood group in human.
- The people who have blood type AB are heterozygous exhibiting phenotypes for both the IA and IB alleles. I isoagglutinin
- In other words, heterozygotes for codominant alleles are phenotypically similar to both parental types.

3. Pleiotropy

- **(from Greek πλείων pleion, 'more', and τρόπος tropos, 'way’)**
- When a single gene affects multiple traits
- Eg sickle cell condition

Pleiotropic disease sickle cell

- Sickle cell anemia is a genetic disease that causes deformed red blood cells with a rigid, crescent shape instead of the normal flexible, round shape.
- It is caused by a change in a gene that encodes information to make the beta-globin subunit of hemoglobin,
- Sickle cell anemia occurs when the gene mutation prevents hemoglobin to transport oxygen.

- Some consequences of the change in the gene:
  - Hemoglobin forms polymers and clump together
  - Red blood cells become rigid
  - HBP
  - Loss of vision

  - That’s one gene, controlling several characters

4. Polygenic inheritance:

- When multiple genes contribute to a single trait.
- polygenetic traits may display a range of possible phenotypes, determined by a number of different genes and the interactions between them.
- The traits that are determined by polygenic inheritance are not simply an effect of dominance and recessively.
- Each of the genes that contributes to a polygenic trait, has an equal influence and each of the alleles has an *additive effect* on the phenotype outcome.

Examples of polygenic traits

- Height
- Intelligence
- Skin colour

Example of multiple genes

- The pigment melanin is responsible for dark coloration in the skin and there are at least three genes, which control for human skin color.
- The production of melanin is controlled by *contributing alleles* (denoted here as A, B and C), resulting in dark skin color,
- Light skin color is produced by *non contributing alleles* (denoted here as a, b and c),

- In two parents, heterozygous for each of the melanin-producing genes (AaBbCc x AaBbCc),
- it is possible to see how the additive effects and combinations of alleles results in all the possible *genotypes*.

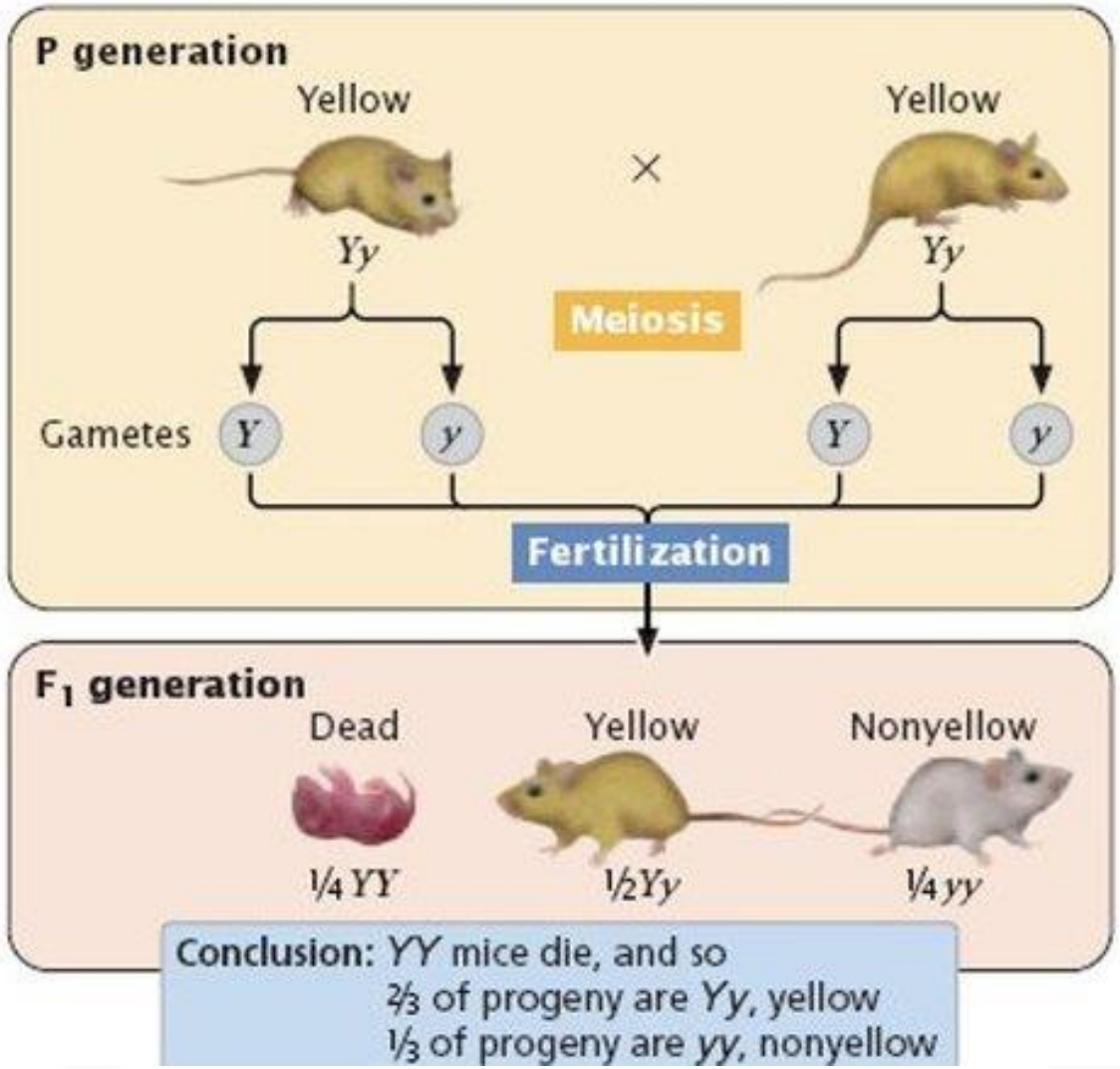
	ABC	*ABc*	*AbC*	*aBC*	*Abc*	*aBc*	*abC*	*abc*
ABC	6	5	5	5	4	4	4	3
ABc	5	4	4	4	3	3	3	2
AbC	5	4	4	4	3	3	3	2
aBC	5	4	4	4	3	3	3	2
Abc	4	3	3	3	2	2	2	1
aBc	4	3	3	3	2	2	2	1
abC	4	3	3	3	2	2	2	1
abc	3	2	2	2	1	1	1	0

6. Gene-environment interaction:

- When environmental factors influence the expression of a gene. Example fur colour in Siamese cats.
- Body parts with temperature below 33°C have black fur while body parts with temperatures above 33°C have white fur.

7. Lethal genes

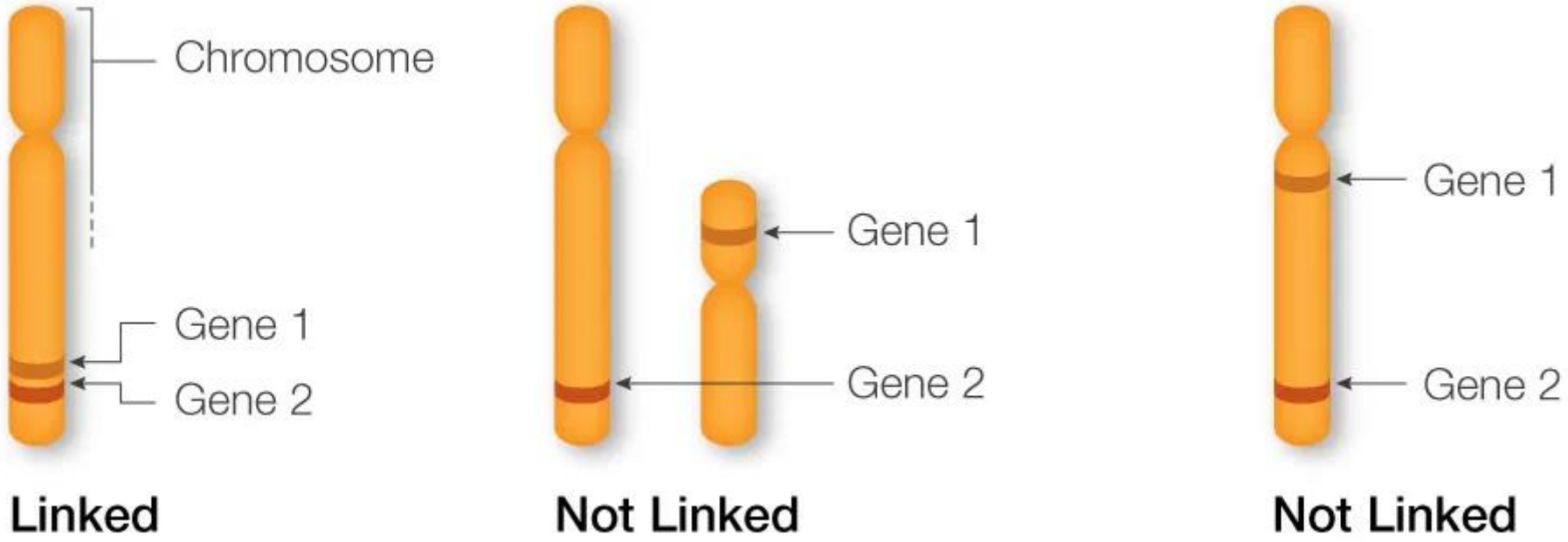
- Gene, which causes the death of its carrier when in homozygous condition is called lethal gene.
- Mendel's findings were based on equal survival of all genotypes.
- In normal segregation ratio of 3:1 is modified into 2:1 ratio
- Example fur colour in mice two alleles one yellow one gray
- Yello dominant to gray. The genotype YY is lethal


P generation
Yellow
Yellow
×
Yy
Yy
Meiosis
Gametes
Y
y
Y
y
Fertilization
F1 generation
Dead
Yellow
Nonyellow
¼ YY
½ Yy
¼ yy
Conclusion: YY mice die, and so
⅔ of progeny are Yy, yellow
⅓ of progeny are yy, nonyellow

Linkage:

- When genes are physically close together on the same chromosome and tend to be inherited together.
- The way in which two genes located close to each other on a chromosome are often inherited together.

Example of linkage



Example of dihybrid cross with linked genes

- Suppose the genes for seed colour and seed shape are linked
- Seed shape Round: Wrinkled (Round dominant to wrinkled)
- Seed colour yellow: green (Yellow dominant to green)
- Draw a cross for heterozygous parents for the two genes

Population genetics

- Population genetics is the branch of genetics that focuses on studying the heritable features of populations over time and space.
- The measurement of variability by describing changes in allele and genotypic frequency for a particular trait over time

Terms

- **1. Population….??**
  - Ecologically: A group of individuals of the same species living within a restricted geographical area that allows any two individuals to interbreed
  - Genetically: A group of individuals who share a common gene pool and have the potential to interbreed

- **2. Gene pool**
  - The sum of all the alleles in a population.
  - A gene pool can also be defined as the alleles among the reproductive members of a population from which gametes can be drawn.

- 3. **The allelic frequency** (or gene frequency)
  - is the rate at which a specific allele appears within a population.
  - Denoted **'P'** for dominant allele and **'q'** for recessive allele
- 4. **Genotypic frequency** : the proportion of individuals in a population that have a specific genotype.

- 5. The genotypic and allelic frequencies constitute genetic structure of a population
- **6. Gene flow:** The movement of genes from one population to another.
- Occurs when individuals of a population migrate.

- 5. Variations: The differences among individuals of a population.
- 6. Morphological Variations: The differences in external appearance among individuals of a population
- 7. Genetic variation: the differences in the DNA sequence among individuals within a population or species

Calculations of genotypic and allelic frequencies

- A frequency is a percentage or a proportion, and always ranges between 0 and 1.
- If 18% of the people in a group have red hair, the frequency of red hair in the group is **0.18** i.e **18÷100 = 0.18**
- To calculate the genotypic frequencies at a specific locus, we count the number of individuals with one particular genotype and divide this number by the total number of individuals in the population.

HARDY-WEINBERG POPULATIONS

- G.H. Hardy and W. Weinberg – developed a law in 1908, which clarifies the relationships between genotype and allele frequencies within a generation and from one generation to another.
- Developed Hardy-Weinberg Law and equation

The Hardy-Weinberg Law

- States that: In a large random mating population with no selection, mutation, or migration, the allelic frequencies and the genotype frequencies are constant from generation to generation

Credibility of the Hardy-Weinberg Law depends on five key points.

- 1. The population includes a very large, virtually infinite number of individuals.
- 2. The individuals mate at random in the sense that each individual's genotype at the locus of interest does not influence his or her choice of a mate.

- 3. No new mutations appear in the gene pool.
- 4. There is no migration into or out of the population.
- 5. There are no genotype-dependent differences in the ability to survive to reproductive age and transmit genes to the next generation.

Hardy-Weinberg equilibrium.

- Populations that satisfy all these five assumptions are said to be at Hardy-Weinberg equilibrium.

- The Hardy-Weinberg equilibrium is a fundamental concept in population genetics that describes the frequency of alleles and genotypes in a population under certain assumptions.

Explanation of the Key Points in Hardy-Weinberg Assumptions

- **1. Random mating**
- The first assumption is random mating, which means that the probability that two genotypes will mate is the product of the frequencies (or probabilities) of the genotypes in the population.
- If the MM genotype makes up 90% of a population, then any individual has a 90% chance (probability = 0.9) of mating with a person with an MM genotype. The probability of an MM by MM mating is (0.9)(0.9), or 0.81

Deviations from random mating come about for two reasons

- **1. choice.**
- If members of a population choose individuals of a particular phenotype as mates more or less often than at random, the population is engaged in assortative mating.

Two types of assortative mating

- If individuals with similar phenotypes are mating more often than at random, positive assortative mating.
- if mating occurs between individuals with dissimilar phenotypes more often than at random, negative assortative mating,

- 2. Circumstance
- Deviations from random mating also arise when mating individuals are either more closely related genetically or more distantly related than individuals chosen at random from the population.
- **Inbreeding** is the mating of related individuals, and outbreeding is the mating of genetically unrelated individuals.

- Inbreeding is a consequence of two factors: **pedigree relatedness** (e.g. cousins) and small population size.

- **2. Large Population Size**
- The second assumption of the Hardy-Weinberg equilibrium is that the population is **infinitely large**.

- A large population produces a large sample of successful gametes.

- The larger the sample, the greater the probability that the allelic frequencies of the offspring will be a true representation of the allelic frequencies in the parental population.

- When populations are small or when alleles are rare, changes in allelic frequencies take place due to chance alone.
- These changes are referred to as random **genetic drift**, or **genetic drift**.

- Allelic and genotypic frequencies may change through the loss or addition of alleles through mutation or migration

- The third and fourth assumptions of the Hardy-Weinberg equilibrium are that neither mutation nor migration causes such allelic loss or addition in the population.

5. No Natural Selection

- The final assumption necessary to the Hardy-Weinberg equilibrium is that no individual will have a reproductive advantage over another individual because of its genotype.
- In other words, no natural selection or artificial selection is occurring.

Hardy-Weinberg equation

- Equation used for calculating genotypic and allelic frequencies in a population

- Assuming that a gene has two allele and each sperm has equal chance of fertilizing any egg.

- Let ‘A’ Denote dominant allele and ‘a’ denote recessive allele
- The frequency for dominant allele is denoted **‘P’** and the frequency for recessive allele is **‘q’**
- Therefore: the frequency for homozygous dominant (AA) is $\mathbf{P^2}$
- The frequency for homozygous recessive (aa) is $\mathbf{q^2}$
- **The frequency for heterozygous (Aa) is 2Pq**

- **This information can be understood from Punnett square when Aa cross with Aa**

The equation

- $P^2 + 2Pq + q^2 + 1$

- Similar: $P + q = 1$

Calculation

- Example 1. Using proportions of earlobe to calculate allelic and phenotypic frequencies

- Example 2.

- A very large population of randomly-mating laboratory mice contains 35% white mice. White coloring is caused by the double recessive genotype, "aa". Calculate allelic and genotypic frequencies for this population.

- Suppose after so many years the population of mice is 100,000. How many mice are homozygous dominant? What assumption do you make for the answer?

- A rather large population of Biology instructors have 396 red-sided individuals and 557 tan-sided individuals. Assume that red is totally recessive. Please calculate the following:

A.The allele frequencies of each allele.

B.The expected genotype frequencies.

C.The number of heterozygous individuals that you would predict to be in this population.

D.The expected phenotype frequencies.

E.Conditions happen to be really good this year for breeding and next year there are 1,245 young "potential" Biology instructors. Assuming that all of the Hardy-Weinberg conditions are met, how many of these would you expect to be red-sided and how many tan-sided?

End of topic

Thank You
for your attention